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CIS6001 - Project proposal

Assignment Title:

Vision-Based Intelligent Assistant for Personal Object Recognition and Location Memory using Deep Learning

Assignment By:

Mohamed Hazari Ahamed Mishal

CMU ID: **20328208**

ICBT ID: **CL/BSCDS/CMU/08/28**

Assessor: **Mr. Roy Ian**

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Abstract

People often misplace their objects like keys, wallets, or phones, which leads to frustration and waste of time. There are existing tracking devices like Apple Air Tag and Tile that help in locating objects that are tagged. However, they cannot visually recognize or remember untagged objects. This research proposes a vision-based intelligent assistant that uses deep learning to detect, track, and remember personal items in real-world environments without the need of any physical tags. The system integrates object detection, object tracking and visual embedding with vector-based memory to create a lightweight, context-aware assistant. Users can query the system through natural language; for example, “Where did I last see my keys?” to find the last known location of an object. Early tests demonstrate the feasibility of integrating memory-based AI and computer vision for routine object retrieval, paving the way for tag-free, human-centered personal memory assistants.

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1. Introduction

People usually misplace important items like keys, wallets, or phones, which often leads to frustration, time loss and reduced productivity. Although commercial trackers such as Apple Air Tag and Tile assist users through Bluetooth or Ultra-Wideband signals. They rely on physical tags and do not provide contextual information about the last location of an object (Lord, 2025). Furthermore, the inability of these systems to visually identify or recall untagged things restricts their usefulness in a variety of real-world settings.

Recent advances in computer vision and deep learning have made it possible for intelligent systems to recognize and understand situations directly from camera input. Modern technologies like YOLOv8 and Vision Transformers (ViTs) have demonstrated remarkable accuracy and real-time efficiency for applications involving object detection and re-identification. (Cheng et al., 2024). Additionally, studies on lightweight models for edge devices have demonstrated that wearable or mobile platforms are capable of high-performance detection (Sun et al., 2025).

In addition to detection, robust multi-object tracking methods like DeepSORT and OC-SORT integrate motion and appearance cues to maintain consistent object identities across frames (Sanyam, 2022). These advancements enable the development of personal assistants that can constantly identify, track, and remember untagged objects in dynamic real-world environments.

Building on this framework, this study proposes a vision-based intelligent assistant that uses a camera to recognize, track, and remember personal objects. Through the integration of YOLOv8 for object identification, DeepSORT for tracking, and a CLIP-based visual memory stored in FAISS. The system could provide descriptive information and/or images in response to common questions such "Where did I last see my wallet?"

This research bridges the gap between memory-augmented personal assistance and visual recognition, advancing context-aware AI and human-centric computing. The ultimate objective is to provide a useful, portable, and privacy-aware solution that enables people to manage their daily possessions without the need for specialized hardware or external tags.

2. Current Situation and Problem Identification

Identity, comfort, and day-to-day functioning are all closely linked to human interactions with personal belongings. A person's feeling of control, order and mental balance may be temporarily disturbed by losing a possession, no matter how small. Because memory and physical surroundings are strongly related, people frequently feel anxious or frustrated when they can't remember where they left an object.(Berry, 2012).

People usually misplace important items like keys, wallets or phones which often leads to frustration, time loss and reduced productivity. Although commercial trackers such as Apple Air Tag and Tile assist users through Bluetooth or Ultra-Wideband signals, they rely on physical tags and do not provide contextual information about the last location of an object (Lord, 2025). Although MIT's RFusion and other cutting-edge research systems use cameras and radio frequency signals to demonstrate AI-assisted retrieval, they require RFID tags and robotic arms which makes them unsuitable for daily personal use. (Zewe, 2021). Additionally, powerful object detection and re-identification capabilities without the need for manual feature creation have been made possible by developments in convolutional neural networks (CNNs), transfer learning and visual embedding approaches (Zhou et al., 2024). These advancements pave the way for a practical, vision-based, context-aware solution that can recognize, store and retrieve untagged personal objects autonomously.

2.1. Problem Statement

“There isn't a vision-based system that can detect, track, and remember untagged personal objects in dynamic real-world environments on its own while providing explainable, context-aware responses to user queries about the last known visual and spatial location of those objects.”

2.2. Aims and Objectives

2.2.1. Aims

To allow explainable, tag-free retrieval of lost items with their last seen visual and geographical context, this project attempts to develop an intelligent, vision-based assistant that can autonomously recognize, label and remember personal objects in a user's environment. For

effective operation on wearable or mobile devices, the system will make use of vector-based memory and lightweight deep learning models.

2.2.2. Research Objectives

1. To design and implement a lightweight vision pipeline integrating real-time object detection, multi-object tracking, and visual embedding for object memory.
2. To develop a vector-based memory module for storing and retrieving visual, temporal, and contextual information of detected objects.
3. To enable natural user queries (e.g., “Where did I last see my keys?”) by retrieving visual and spatial evidence from stored memory.
4. To evaluate the system’s performance in dynamic, moving camera scenarios in terms of accuracy, efficiency, and user interpretability.

2.3. **Research Questions**

1. How can a lightweight vision-based system effectively detect, track, and remember untagged personal objects in real-world dynamic environments?
2. What visual embedding and memory mechanisms can ensure reliable, long-term object identification and retrieval?
3. How can stored visual and spatial data be efficiently searched and presented in response to natural user queries?
4. How effective is the proposed system in delivering accurate, explainable, and context-aware responses under varying environmental and camera conditions?

2.4. **Research Gap**

Modern consumer tracking systems lack contextual or visual memory and are dependent on external hardware tags. Though they still rely on tags and robotic manipulation, research prototypes like RFusion show that merging vision and RF data is feasible. On the other hand, vision-memory and SLAM-based methods frequently require regulated settings or substantial processing power, which restricts their use for wearable or portable devices. For everyday personal object management, there is still a need for a tag-free, lightweight, context-aware system that combines visual memory, real-time detection, tracking, and re-identification.

3. Literature Review

Research on personal items that go missing has always been an issue, however, the researchers have tracked, AI and memory-augmented systems to come up with different solutions. Among them were the traditional and more sophisticated methods of UWB/Bluetooth fusion (Li et al., 2024) and BLE/RFID-based asset tracking (Bisio et al., 2016) which indicate that it is possible to localize indoors with fair accuracy. Still, these methods have limitations as they are based on the use of physical tags or fixed infrastructure and do not provide much visual context hence less applicable for the case of personal objects every day.

There have been some mixed works that relate tagging to a vision concept, such as the RFID-empowered robotic systems (Zewe, 2021). Since such systems can even find objects that are out of their sight, they require the presence of physical tags and robotic systems and, therefore, cannot be practical for everyday personal use. Meanwhile, advances in deep learning have paved the way for more flexible solutions. CNN-based object recognition (Tsirtsakis et al., 2025) and Transformer-based re-identification models (Ye et al., 2024) demonstrate that computers can robustly learn different features and deliver excellent results in tracking object locations night and day, which is an important element to establish when a lost item was last seen.

Techniques from SLAM and visual SLAM research (Xin and Li-Na, 2022) add another layer by providing spatial context, allowing a system to map where objects appear within a scene. Multimodal embeddings, like CLIP (OpenAI, 2022), even allow integration of visual and textual information, enabling natural language queries to find objects efficiently. Lightweight architectures such as LostNet (Zhou et al., 2024) prove that accurate image recognition and retrieval can happen on resource-limited devices, which is essential for mobile or wearable implementations.

Overall, this research demonstrates advancements in tracking, recognition and memory systems while also highlighting the lack of a workable, vision-based, context-aware solution that can assist individuals in remembering everyday items without the use of tags or specific hardware. This gap drives the development of a personal assistant that can recognize, follow and recall objects in natural settings, providing a natural and understandable response to queries like "Where did I last see my keys?"

3.1. Summary of Existing Work

Representative Studies	Key Techniques / Systems	Strengths	Limitations	Relevance
UWB + Bluetooth Fusion (Li et al., 2024)	Kalman filter + particle fusion for indoor localization	Achieves < 1 m accuracy in indoor tracking	Needs infrastructure (beacons/tags); no visual context	Strong comparative baseline; supports need for tag-free vision approach
BLE + RFID Asset Tracking (Bisio et al., 2016)	Smartphones + BLE/RFID tags, GPS, energy-optimized algorithms	Demonstrates practical asset tracking with long smartphone battery life and ~2 m accuracy	Requires physical tags; works best in structured environments; lacks visual/contextual memory	Illustrates limitations of tag-based tracking; motivates need for vision-based, tag-free personal object tracking
RFID + Vision (Zewe, 2021)	RF signals + robotic vision for object retrieval	Demonstrates AI-assisted search capability	Depends on RFID tags and robot hardware impractical for everyday use	Shows potential of AI fusion but still tag-dependent
Deep Learning for Object Recognition (Tsirtsakis et al., 2025)	CNNs, DL algorithms for classification, localization, and detection	Comprehensive survey of object recognition methods, datasets, and benchmarks	Focused on general object recognition; lacks multi-object tracking and memory for everyday personal items	Provides foundational understanding for vision-based object detection and tracking in personal environments
Transformer-based Object	Vision Transformers	Comprehensive survey of	Focused on Re-ID datasets and	Shows the state-of-the-art in using

Re-Identification (Ye et al., 2024)	(ViT), UntransReID, unsupervised Re-ID	Transformer-based Re-ID methods; introduces unsupervised Re-ID baseline; explores cross-modal, animal, and special Re-ID scenarios	benchmarks; less on integration with real-world personal object tracking	Transformer architectures for object Re-ID, useful for improving multi-object tracking and recognition in personal environments
SLAM and vSLAM for Mobile Robots (Xin and Li-Na, 2022)	Feature-based (ORB, SIFT, SURF), Direct methods, EKF/UKF/Particle filters, RGB-D and stereo vision	Enables autonomous robot localization and mapping in unknown environments; supports single and cooperative multi-robot exploration; feature extraction, keyframe mapping, and dense/semi-dense 3D reconstruction	High algorithmic complexity; computationally intensive; sensitive to dynamic or low-texture environments; challenges in real-time and large-scale mapping	Provides foundational understanding of visual-based localization and mapping techniques, informing design of mobile/wearable systems for object tracking, environment mapping, and memory-based retrieval
CLIP (Contrastive Language Image Pretraining (OpenAI, 2022)	Joint image-text embedding using contrastive pretraining; Transformer-	Enables learning of visual concepts from natural language supervision; supports zero-shot	Requires massive web-scale image-text datasets; high computational requirements;	Serves as a foundation for multimodal, natural-language-guided visual recognition;

	based image and text encoders; zero-shot classification	classification across multiple datasets; improves robustness across diverse domains; reduces reliance on expensive labeled datasets	may underperform in specialized domains not covered in training	informs design of systems that leverage text to guide flexible image understanding
LostNet(Zhou et al., 2024)	MobileNetV2-based CNN with Convolutional Block Attention Module (CBAM), perceptual hashing, transfer learning; Spring Boot backend for online service	Provides an efficient online lost and found image recognition system; achieves 96.8% testing accuracy with low computational cost (0.67 GFLOPs, 3.5M parameters); integrates frontend-backend segregation for maintainability	Dataset limited to curated sources; performance may vary with unstructured real-world data; scalability to extremely large datasets not fully tested	Demonstrates a lightweight, high-accuracy image-matching system suitable for real-time lost-and-found applications; informs design of efficient multimodal recognition systems

Table 1 - Literature Review

4. Proposed Solution

The system that is being proposed is an intelligent personal assistant based on vision, which allows the users to recognize, follow and remember everyday objects in real-world environments without requiring physical tags or labels. The system, using mobile or wearable cameras together with a lightweight backend, creates a context-aware visual memory that can keep and retrieve information about personal items in a visual and spatial manner. This memory allows the user to query for objects' last known locations through natural language queries.

4.1. Methodological Overview

The system is designed around five integrated modules:

1. Object Detection:

- Utilize YOLO model for real-time, lightweight object detection, optimized for mobile and low-power GPU environments.
- Detects multiple objects per frame and outputs bounding boxes with confidence scores.

2. Object Tracking:

- Implement a suitable model such as DeepSORT or StrongSORT to maintain consistent object IDs across frames.
- Aggregates multi-view evidence, enabling reliable short-term tracking even under occlusions or temporary exits from the camera frame.

3. Embedding and Re-Identification:

- Generates object embeddings using CLIP or a custom CNN head for appearance-based long-term recognition.
- Allows robust object re-identification across different time frames and perspectives.

4. Memory and Label Association:

- Users can assign semantic labels (e.g., “Scooty key” or “Wallet”) to objects.

- Memory entries include embeddings, timestamps, frame snapshots, and approximate camera pose or scene tags.
- Entries are stored in a vector database (FAISS/Chroma) with a relational metadata index for efficient retrieval.

5. Query Interface:

- Natural language queries (e.g., “Where did I last see my keys?”) retrieve the last seen visual and spatial context.

This multi-layered system integrates the traditional approaches of detection/tracking with the unconstrained use of personal cameras, with a strong focus on reliable object re-identification, even when there are a lot of camera movement and the environment changes.

5. Feasibility Analysis

1. Technical Feasibility:

- All components (YOLO, DeepSORT/StrongSORT, CLIP, FAISS, mobile APIs) are open-source and well-documented.
- The prototype can be implemented within the semester using standard computing resources.

2. Economic Feasibility:

- The system requires only common laptops, smartphones, or wearable cameras.
- No proprietary hardware is needed, minimizing cost and enabling scalability.

3. Operational Feasibility:

- Local storage and opt-in labeling maintain user privacy and control.
- Edge/cloud split and model optimization (pruning, quantization) ensure efficient real-time operation on resource-constrained devices.

6. Project Description

The system allows for the intuitive retrieval of personal things by combining vector-based memory, visual embedding for re-identification, multi-object tracking, and real-time detection. The ability to query for an object's last observed visual and geographical environment offers both usefulness and a proof-of-concept illustration of an object memory system without tags.

Prototype Evaluation:

- Testing will be conducted using handheld or wearable cameras in dynamic environments such as homes, offices, and outdoor areas.
- Evaluation will focus on accuracy, efficiency, and usability, simulating realistic daily scenarios.

7. Deliverables

1. Prototype Application:

- Mobile/wearable application capturing live video and performing detection, tracking, and memory storage.

2. Re-Identification Module:

- Visual embedding module integrated with vector-based memory, enabling label association and long-term object recognition.

3. Query Interface:

- Natural language interface for retrieving last seen visual and spatial context.

4. Documentation:

- System architecture, algorithm description, usage guide, evaluation results, and future improvement recommendations.

8. Resources Required

1. Hardware:

- Development laptops or desktops with moderate GPU support (e.g., NVIDIA RTX series) for model training and testing.
- Smartphones (Android or iOS) or wearable cameras for live video capture and prototype testing.

2. Software & Libraries:

- Open-source frameworks and libraries including Ultralytics YOLOv8 for object detection, DeepSORT/StrongSORT for tracking, CLIP for visual embeddings, FAISS for vector-based memory search, and SQLite/Postgres for metadata storage.
- Python-based environment for machine learning and backend modules.

3. Other Tools:

- Lightweight SLAM or visual odometry libraries (e.g., ORB-SLAM variants) to provide camera pose and scene context for enhanced memory retrieval.
- Cloud storage or local storage infrastructure for videos, embeddings, and associated metadata.

9. Expected Output and Outcome

Expected Output:

- Functional prototype demonstrating real-time detection, tracking, memory, and query retrieval.
- Accurate label-to-object embedding association for multiple small items.
- On-demand retrieval of last seen visual and spatial context.

Expected Outcome:

- Reduced time and frustration in locating misplaced items.
- A tag-free, practical system for everyday personal use.
- Insights into robust object re-identification under dynamic, unconstrained camera conditions.

10. Time Plan for Implementation

Task	Description	Nov 2025	Dec 2025	Jan 2026	Feb 2026
Literature Review & Finalization	Review latest papers on vision-based tracking, memory models, and re-identification systems	Conduct and finalize	—	—	—
Dataset Preparation	Collect/curate object image dataset and perform annotation	Initiate	Complete	—	—
Model Selection & Setup	Configure YOLOv8 and CLIP environments; test sample runs	Commence	Finalize	—	—
Object Detection & Tracking Implementation	Integrate YOLOv8 + DeepSORT; validate multi-object tracking	—	Commence	Finalize	Complete
Embedding & Vector Memory Module	Build FAISS-based vector storage for embeddings	—	Implement	Refine	—

Query Interface Development	Implement natural language interface (text + speech)	—	Initiate	Complete	—
System Integration & Testing	Combine modules, debug, and perform scenario testing	—	Partial integration	Unit/Integration testing	Full Testing
Evaluation & Refinement	Conduct accuracy and performance tests under real-world conditions	—	—	Conduct	Complete
Documentation & Presentation	Prepare final report, documentation, and presentation materials	—	—	Commence	Finalize

Table 2 - Time Plan for Implementation

11. Limitations, Risks, and Critical Evaluation

Limitations:

- Performance may degrade under extreme lighting, rapid motion or occluded objects.
- Memory database size may constrain the number of objects stored.

Risks:

- Mobile/wearable hardware limitations could affect speed and reliability.
- Privacy concerns regarding continuous video capture.
- User-provided labels may be inconsistent or inaccurate.

Critical Evaluation:

- This project emphasizes proof-of-concept feasibility rather than large-scale deployment.
- Success metrics include detection/tracking accuracy, memory retrieval reliability and usability in real-world scenarios.
- Optimization strategies (model pruning, quantization, and efficient vector search) are critical for resource-constrained deployment.

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